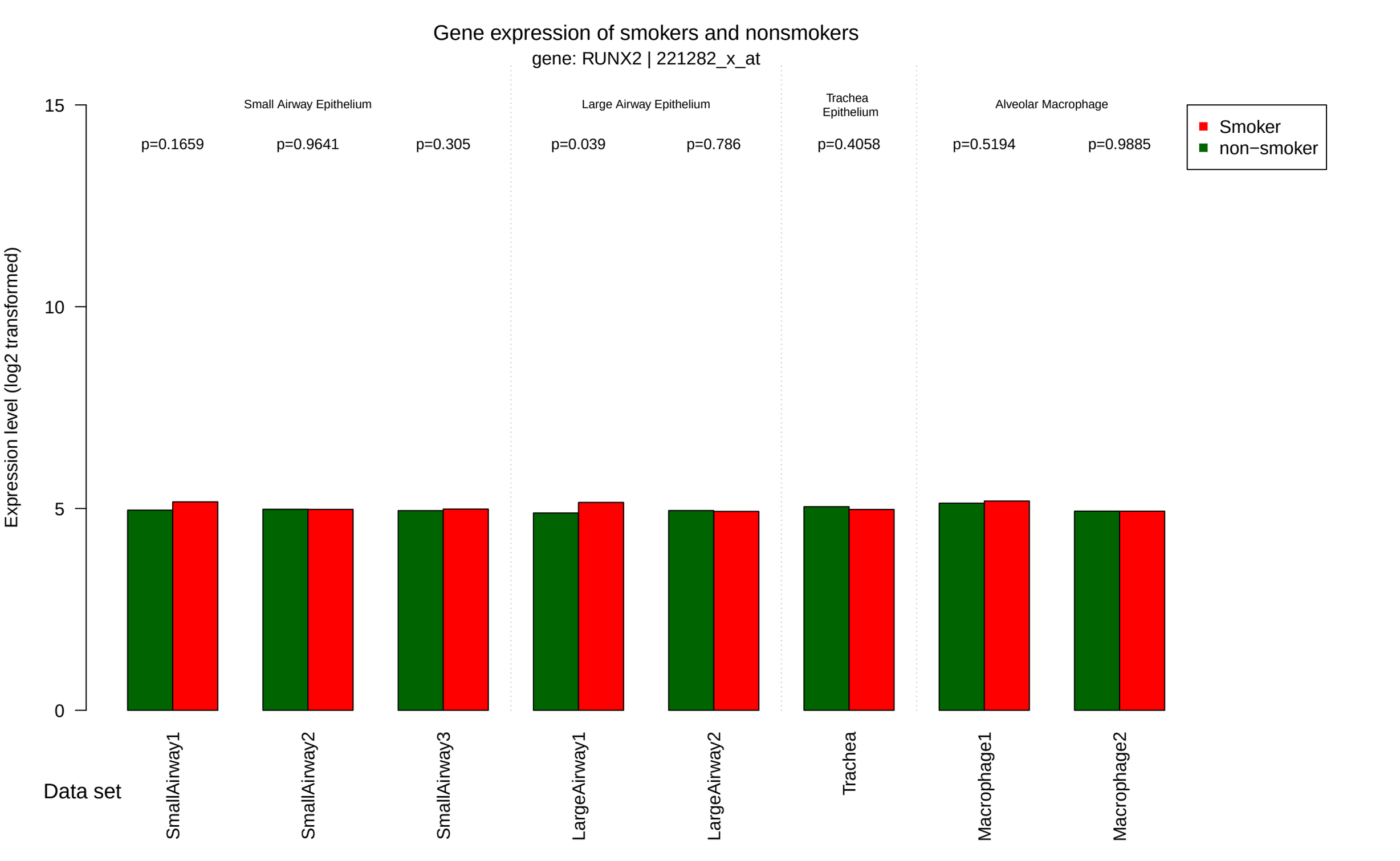
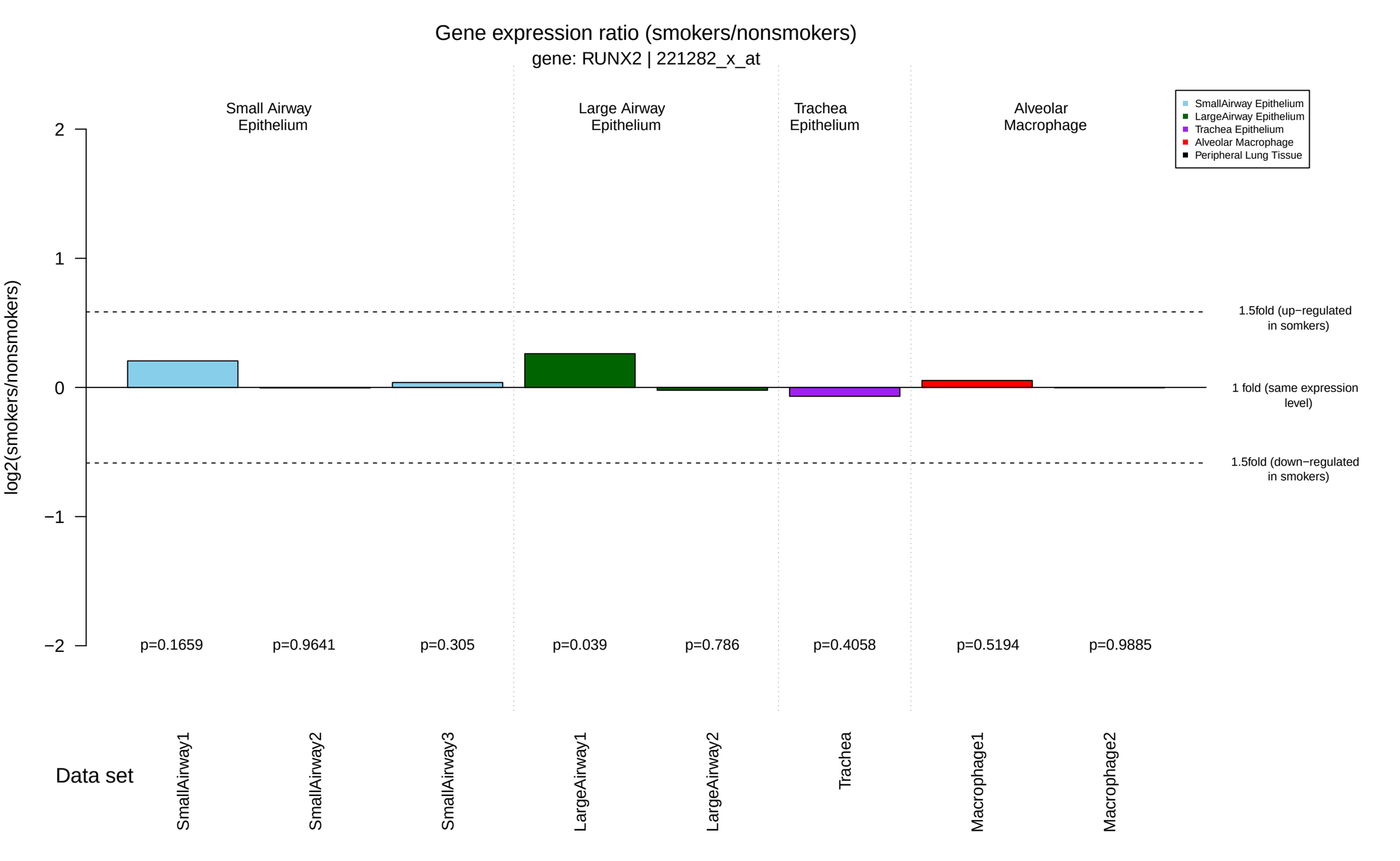
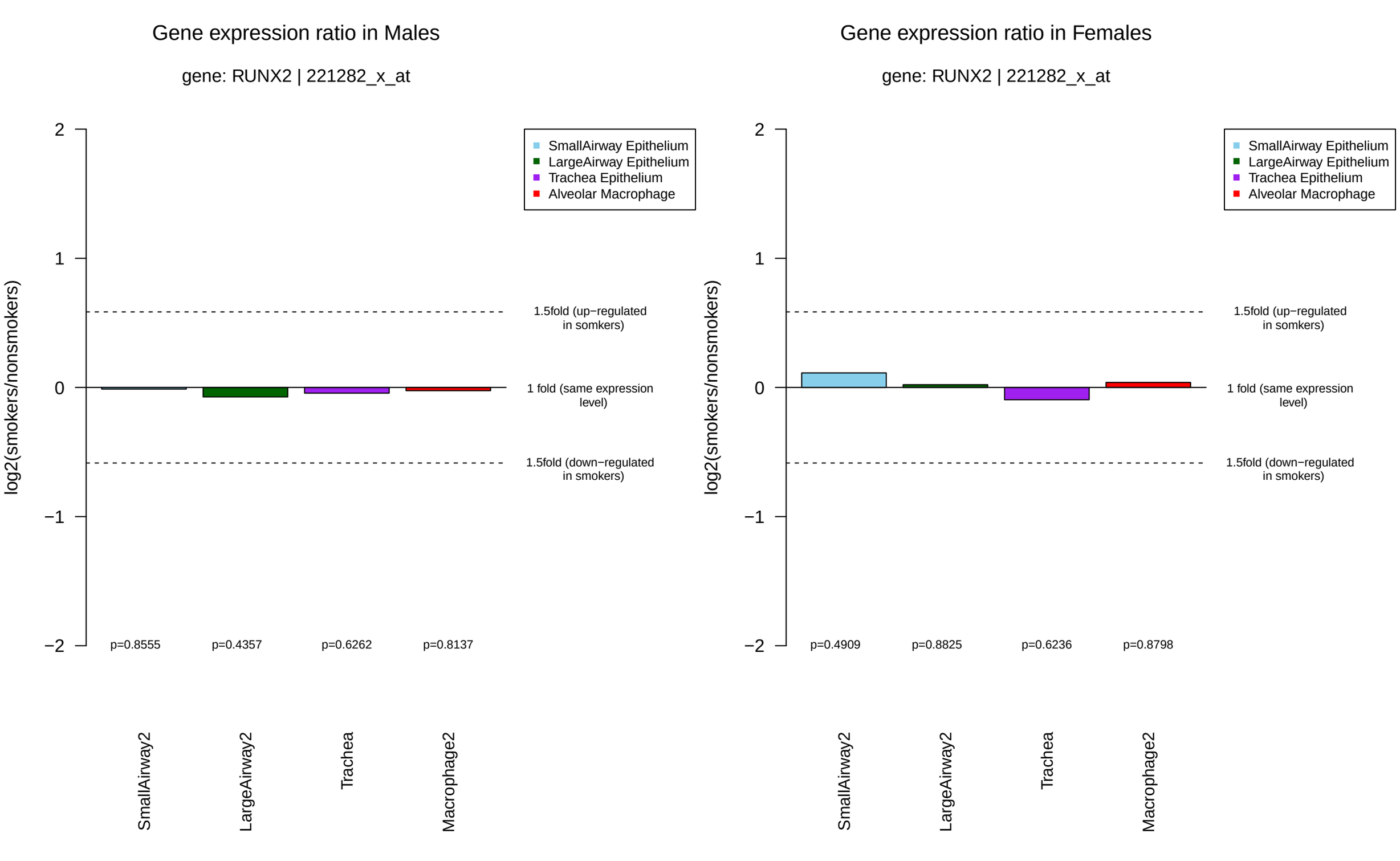
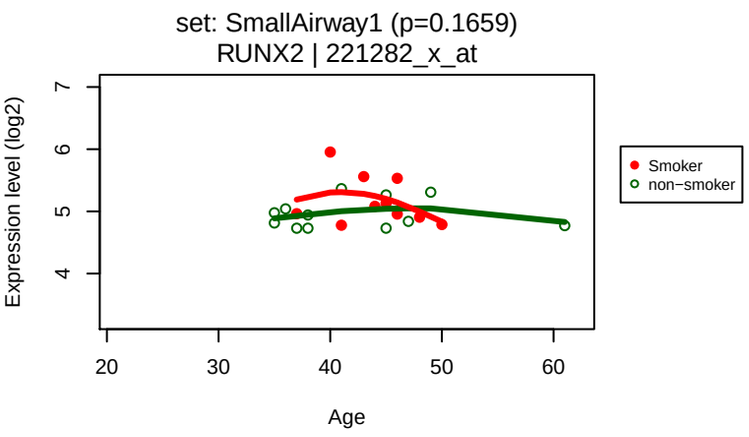
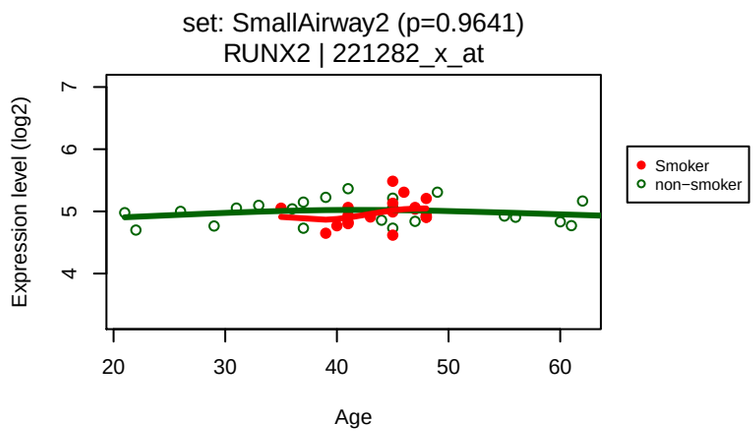
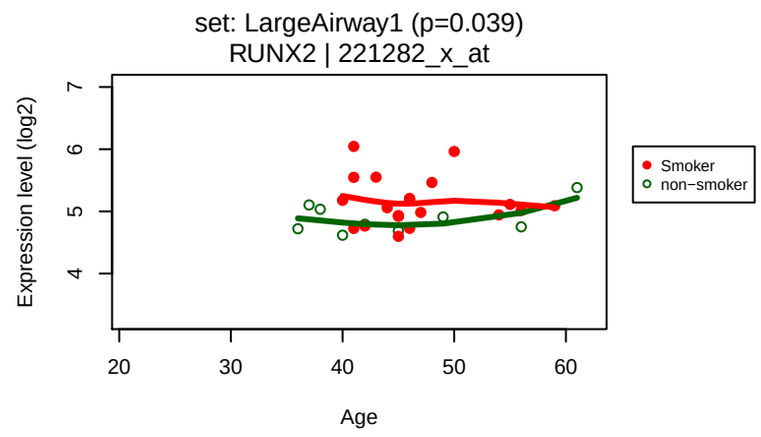

Gene expression of smokers and nonsmokers
gene: RUNX2 | 221282_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1659
p=0.9641
p=0.305
p=0.039
p=0.786
p=0.4058
p=0.5194
p=0.9885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

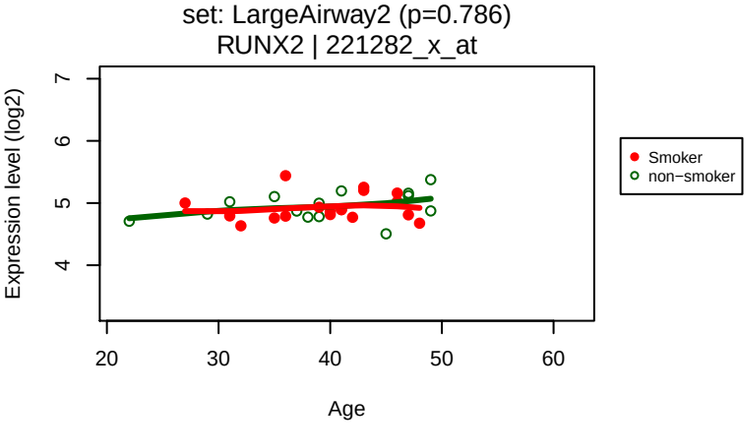

Gene expression ratio (smokers/nonsmokers)
gene: RUNX2 | 221282_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1659
p=0.9641
p=0.305
p=0.039
p=0.786
p=0.4058
p=0.5194
p=0.9885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

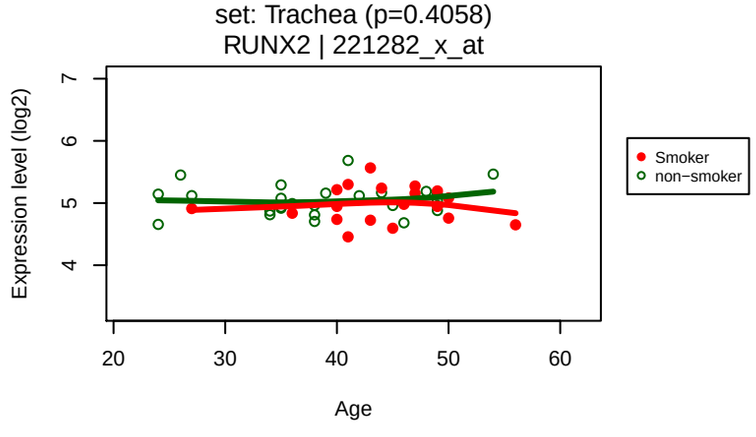

Gene expression ratio in Males
gene: RUNX2 | 221282_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8555
p=0.4357
p=0.6262
p=0.8137
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX2 | 221282_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4909
p=0.8825
p=0.6236
p=0.8798
SmallAirway2
LargeAirway2
Trachea
Macrophage2

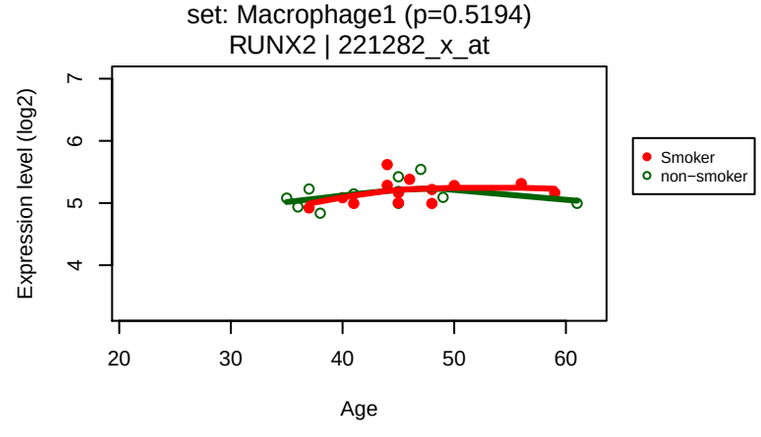

set: SmallAirway1 (p=0.1659)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker

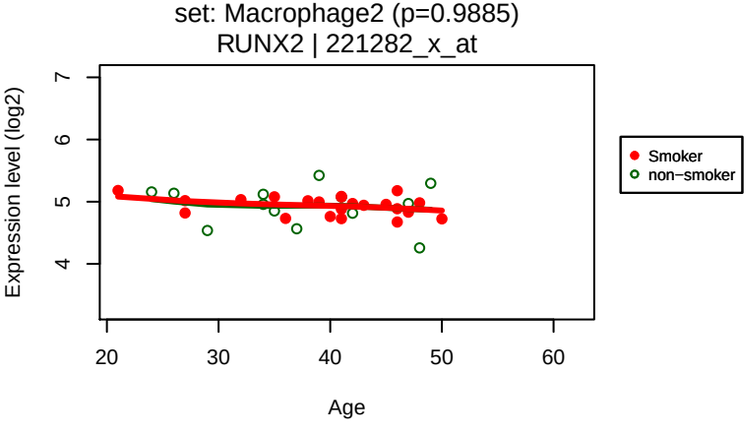

set: SmallAirway2 (p=0.9641)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker

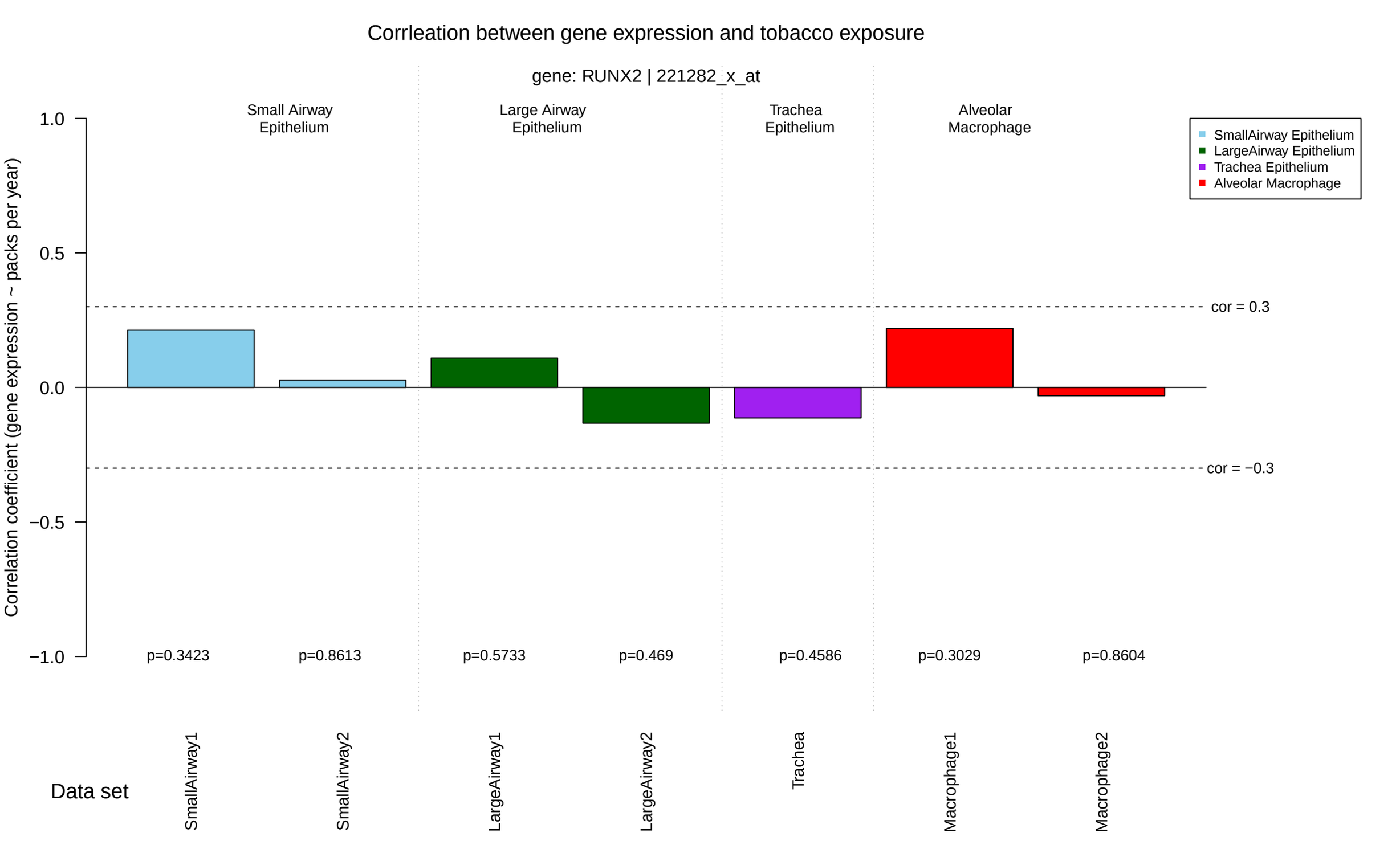

set: LargeAirway1 (p=0.039)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.786)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4058)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5194)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9885)
RUNX2 | 221282_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX2 | 221282_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3423
p=0.8613
p=0.5733
p=0.469
p=0.4586
p=0.3029
p=0.8604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2